

B.Tech DEGREE EXAMINATION, JANUARY 2024

First Semester

21PYB102J - SEMICONDUCTOR PHYSICS AND COMPUTATIONAL METHODS*(For the candidates admitted during the academic year 2022-2023 onwards)***Note:**

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours**Max. Marks: 75****PART - A (20 × 1 = 20 Marks)**

Answer all Questions

Marks BL CO

- | | | | |
|---|---|---|---|
| 1. Quantum free electron theory was proposed by..... | 1 | 1 | 1 |
| (A) Bloch | | | |
| (B) Drude & Lorentz | | | |
| (C) Sommerfeld | | | |
| (D) Newton | | | |
| 2. The probability that an electron in a metal occupies the Fermi-level, at any temperature above 0 K is..... | 1 | 2 | 1 |
| (A) 0 | | | |
| (B) 0.5 | | | |
| (C) 1 | | | |
| (D) 100 | | | |
| 3. A semiconductor behave like a _____ at absolute zero temperature. | 1 | 1 | 1 |
| (A) Conductor | | | |
| (B) Insulator | | | |
| (C) Protection device | | | |
| (D) Quasi conductor | | | |
| 4. _____ the eigen function of d/dx. | 1 | 1 | 1 |
| (A) X^2 | | | |
| (B) $2 \sin(x)$ | | | |
| (C) e^{2x} | | | |
| (D) $\sin(2x)$ | | | |
| 5. Germanium is the example of semiconductor | 1 | 1 | 2 |
| (A) Intrinsic | | | |
| (B) Extrinsic | | | |
| (C) Metal Doped | | | |
| (D) Oxide Doped | | | |
| 6. The photo diode accepts.....energy as input to generate..... | 1 | 2 | 2 |
| (A) Sound, Electric current | | | |
| (B) Heat, Electric current | | | |
| (C) Electric current, Light | | | |
| (D) Light, Electric current | | | |
| 7. In a semiconductor the drift current is due to | 1 | 2 | 2 |
| (A) Random motion of electrons in the absence of electric field | | | |
| (B) Random motion of holes in the absence of electric field | | | |
| (C) Movement of charge carriers in the presence of applied electric field. | | | |
| (D) Movement of the positive ions in the presence of applied electric field. | | | |
| 8. converts electrical energy into light energy. | 1 | 2 | 2 |
| (A) Solar Cell | | | |
| (B) Light Emitting Diode | | | |
| (C) Dynamo | | | |
| (D) Liquid crystal display | | | |
| 9. Band to Band transition in GaAs can results absorption and emission of photons with wavelength of..... | 1 | 2 | 3 |
| (A) 0.087 nm | | | |
| (B) 0.087 mm | | | |
| (C) 0.87 μm | | | |
| (D) 0.087 $\text{\AA}$ | | | |
| 10. Spontaneous emission processes is directly proportional to..... | 1 | 2 | 3 |
| (A) Number of atoms in excited state | | | |
| (B) Number of atoms in ground state | | | |
| (C) Energy density of radiation | | | |
| (D) Number of holes in ground state | | | |

- | | | | |
|--|---|---|---|
| 11.rule is a simple expression for the transition probabilities between states of a quantum system, which are subjected to a perturbation | 1 | 2 | 3 |
| (A) Ampere | | | |
| (B) Maxwell | | | |
| (C) Einstein | | | |
| (D) Fermi Golden | | | |
| 12.is a popular simulation technique for evaluating the evolution of optical fields in waveguides and photonic devices, and Optoelectronic Integrated Circuits | 1 | 2 | 3 |
| (A) Beam Propagation Method | | | |
| (B) Transfer Matrix Method | | | |
| (C) Pseudo Potential Method | | | |
| (D) Density Function Method | | | |
| 13. The method is used to measure the resistance | 1 | 1 | 4 |
| (A) Hydrogenation | | | |
| (B) Electrolysis | | | |
| (C) Rectification | | | |
| (D) Vander Pauw | | | |
| 14. C-V measurements are capable of yielding information about the and concentration of charge carriers | 1 | 2 | 4 |
| (A) Drift potential | | | |
| (B) Diffusion potential | | | |
| (C) Bonding | | | |
| (D) Structure | | | |
| 15. In four probe method, the resistance due to the superposition of current at outer tips R =..... | 1 | 1 | 4 |
| (A) $V/2I$ | | | |
| (B) V/I | | | |
| (C) $2V/I$ | | | |
| (D) $2VI$ | | | |
| 16. The resistivity of the given material is _____, whose resistance is 2Ω ; area of cross-section and length are 25 cm^2 and 15 cm respectively. | 1 | 3 | 4 |
| (A) 333 Ohm m | | | |
| (B) 3.33 Ohm m | | | |
| (C) 0.0333 Ohm m | | | |
| (D) 0.333 Ohm m | | | |
| 17. Nano scale materials has their size in between.....nm | 1 | 1 | 5 |
| (A) 10-100 | | | |
| (B) 100-1000 | | | |
| (C) 1000-2000 | | | |
| (D) 1-100 | | | |
| 18. In a quantum dot the electron confinement is along _____. | 1 | 2 | 5 |
| (A) One dimension | | | |
| (B) all the three dimensions | | | |
| (C) Two dimensions | | | |
| (D) zero dimension | | | |
| 19. In CVD process , the precursor in the form of..... | 1 | 1 | 5 |
| (A) Liquid | | | |
| (B) Solid | | | |
| (C) Gas | | | |
| (D) Semi solid | | | |
| 20. technique allows surfaces of objects to be seen in their natural state without staining | 1 | 2 | 5 |
| (A) XEM | | | |
| (B) AFM | | | |
| (C) TEM | | | |
| (D) XRD | | | |

PART - B ($5 \times 8 = 40$ Marks)

Marks BL CO

Answer all Questions

- | | | | |
|---|---|---|---|
| 21. (a) Illustrate the importance of density of State and obtain the expression for density of state. | 8 | 3 | 1 |
| (OR) | | | |
| (b) (i) What is meant by Brillouin Zone? Draw the diagram for Brillouin zone in 1D and 2D Lattice. (4 Marks) | | | |
| (ii) Explain the concept of Eigen function, Eigen value and Eigen equation through an example of function. (4 Marks) | | | |
| 22. (a) Illustrate the variation of Fermi level with temperature for N-Type and P-Type semiconductor. | 8 | 4 | 2 |
| (OR) | | | |
| (b) Outline the process of diffusion involved in the formation of a PN junction. Derive an expression for continuity equation governing the transport of mobile charge carriers in a semiconductor. | | | |

23. (a) Define Density of states for photons. Obtain the expression for density of state for photons in-terms of energy and frequency. 8 4 3
(OR)
(b) (i) Illustrate the concept of Fermi Golden rule. Obtain the expression for transition rate per unit volume of the system. (5 Marks)
(iii) Determine the conversion efficiency, η of the solar cell, if Short-Circuit current = 2.8 A, Open-Circuit Voltage = 0.55V, Fill Factor (FF) = 0.8 and Input Power (P_{in}) = 10 W. (3 Marks)
24. (a) Examine how the Hall Coefficient of a given semiconductor is related to Hall voltage by deriving a relation between them. 8 3 4
(OR)
(b) (i) Explain the working concept of Hot point probe method. (4 Marks)
(ii) What is Monte Carlo method? Explain the estimation of Pi value Monte Carlo method. (4 Marks)
25. (a) Derive an expression the Density of States in 2D, 1D and 0D. 8 4 5
(OR)
(b) Illustrate the construction and working principle of Physical Vapour Deposition (PVD).

PART - C ($1 \times 15 = 15$ Marks)

Marks BL CO

Answer **any 1** Questions

26. (i) Apply Bloch theorem and derive the expression for an electron in a periodic field of 1D crystal Lattice. (12 Marks) 15 4 1
(ii) Discuss the energy of electron under the condition $P \rightarrow 0$. (3 Marks)
27. (i) Illustrate with a neat sketch the construction and working principle of Transmission Electron Microscope (TEM). Comment on the interaction of the electron beam with the sample in SEM and TEM techniques. (10 Marks) 15 4 5
(ii) Describe the working concept of Atomic Force Microscope with neat diagram. (5 Marks)

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SRM Institute of Science and Technology
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DEPARTMENT OF PHYSICS
B.Tech/M.Tech Integrated DEGREE EXAMINATION

Faculty Name: Dr. A. Senthil

FacultyID: T239

Course Code: 21PYB102J

Course Name: Semiconductor Physics and Computational Methods

Date of Exam: 06/01/2024

Duration: 3 Hours

PART—A (20x1=20 Marks)
Answer ALL Questions

1. (C) Sommerfeld
2. (B) 0.5
3. (B) Insulator
4. (C) e^{2x}
5. (A) Intrinsic
6. (D) Light, Electric current.
7. (C) Movement of charge carriers in the presence of an applied electric field.
8. (B) Light Emitting Diode
9. (C) $0.87 \mu\text{m}$
10. (A) Number of atoms in excited state
11. (D) Fermi's Golden Rule
12. (A) Beam Propagation Method
13. (D) Vander Pauw method
14. (B) Diffusion potential
15. (B) V/I
16. (D) 0.333 ohm m
17. (D) 1-100 nm
18. (B) All three dimensions
19. (C) Gas
20. (B) AFM

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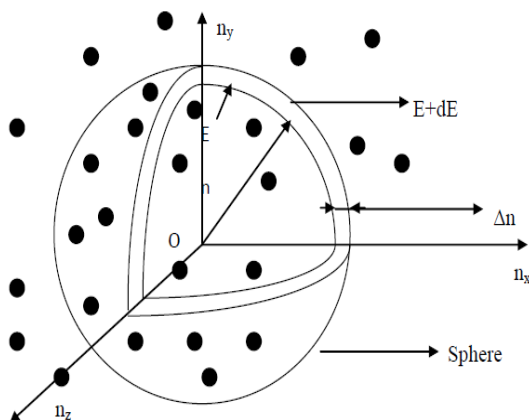
Duration: 3 Hours

PART— B(5 x 8= 40 Marks)

Answer ALL Questions

Q-21 (a) Definition- Diagram-Importance of density of State - 2 marks

Definition: Density of States $Z(E) dE$ is defined as the number of available electron states per unit volume in an energy interval (dE).



Derivation – 6 Marks

$$Z(E)dE = \pi / 4 \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE \longrightarrow (5)$$

Since each energy level provides 2 electron states one with spin up and another spin down, we have

$$\begin{aligned} \text{Density of states } Z(E)dE &= 2 \times \pi / 4 \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE \\ Z(E)dE &= \pi / 2 \left[\frac{8m}{h^2} \right]^{3/2} E^{1/2} dE \longrightarrow (6) \end{aligned}$$

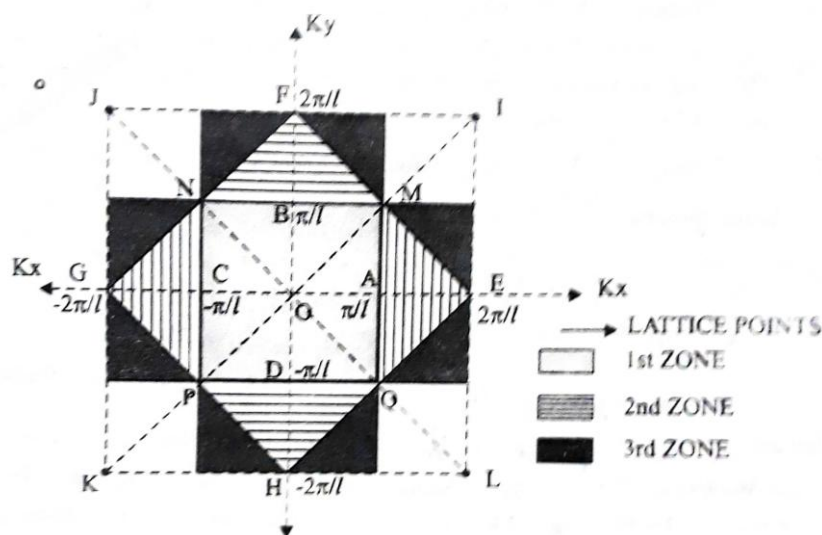
(OR)

(b) (i) Explanation of Brillouin zone – 2 marks;

Brillouin Zones:

Brillouin zones are the boundaries marked by the values of $\vec{k}$ in which electrons can have allowed energy values without diffraction.

Diagram of Brillouin zone 1D,2D Lattice- 2 marks



(ii) Concept of Eigen function, Eigen value and Eigen equations : 3marks ;

Examples: 1mark

In quantum mechanics, if we apply an operator on wave function then the same wave function is achieved back with a scalar value. In that scenario, the Wave function is called Eigen function and the scalar value is called Eigenvalue. An equation that includes Eigen function, operator and Eigen value is called the Eigen equation.

$$\frac{d \sin(x)}{dx} = -\cos(x)$$

$$\frac{d e^{3x}}{dx} = 3e^{3x}$$

No same function is achieved back after applying differential operator, (i.e. Sin and Cos functions are different). Where 3 is eigen value The same function is achieved back with a multiple of '3' after applying the differential operator.

Q.22 (a). Variation of Fermi level with temperature for N- type (4 marks) ;

The Fermi level for N-type semiconductor is given by

$$E_F = \frac{E_c + E_d}{2} + \frac{kT}{2} \ln \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* kT}{h^2} \right)^{3/2}} \right] \rightarrow (1)$$

When Temperature $T = 0K$, eqn (1) becomes

$$E_F = \frac{E_c + E_d}{2} \rightarrow (2)$$

- At $T=0K$, The Fermi level lies exactly half way between E_c and E_d .
- When temperature is increased, the donor atoms are ionised and the contribution of electrons from E_d to E_c increases.
- This results the Fermi level E_F shifts down as shown in figure.
- Further increase in temperature results in generation of electron-hole pair due to the breaking of covalent bonds and the material behave like in intrinsic manner. (7)

Variation of Fermi level with temperature for P- type (4 marks)

The Fermi level for P-type semiconductor is given by

$$E_F = \frac{E_v + E_a}{2} - \frac{kT}{2} \ln \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* kT}{h^2} \right)^{3/2}} \right] \rightarrow (1)$$

When Temperature $T = 0K$, eqn (1) becomes

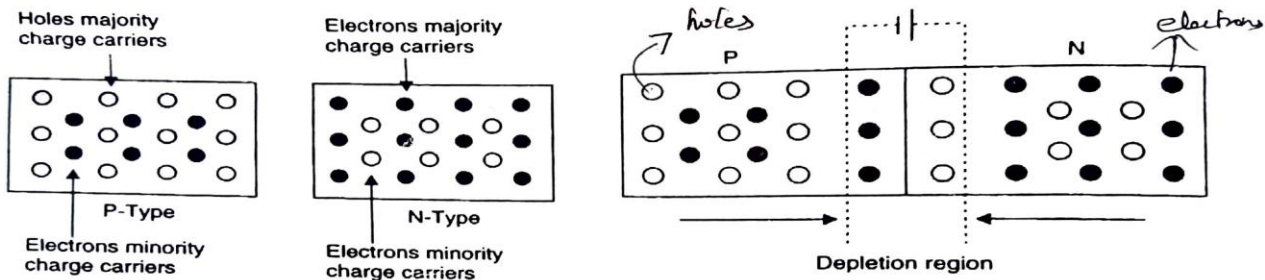
$$E_F = \frac{E_v + E_a}{2} \rightarrow (2)$$

- At $T=0K$, the Fermi level lies exactly half way between E_v and E_a .
- When temperature is increased, the acceptor atoms are ionized and the contribution of electrons from E_v to E_a increases.

(OR)

(b) Processes of diffusion involved in formation of a PN junction – 3

- When two semiconducting materials, p-type and n-type are brought into contact, the majority carrier of each type would diffuse across the junction.
- The diffusion would stop after an electric field is built up sufficiently high to oppose diffusion.
- As the majority carrier such as hole diffuses across the junction, it combines with electron in the n-type side, which creates a net positive charge.



- Likewise, the majority carrier electron from n-type material diffuses across the junction recombines with hole in p-type side creates net negative charge.
- The net charge at each side creates an electric field in the direction, which would oppose further diffusion. This region is called as diffusion region.
- The electric field created would drift the minority carrier in the opposite direction across the junction.
- Thus when equilibrium attained, the drift carriers and diffused carriers should be balanced.

Continuity equation for transport of mobile charge carriers - 5 marks

Definition: 1 mark

Equation with explanation: 4 marks

The contribution of the overall effect when drift, diffusion, and recombination occur simultaneously in a semiconductor material. The governing equation is called the continuity equation

The continuity equation describes a basic concept, namely that a change in carrier density over time is due to the difference between the incoming and outgoing flux of carriers plus the generation and minus the recombination.

Consider an infinitesimal slice with thickness dx located at x . The number of electrons in the slice may increase due to the net current flow into the slice and the net carrier generation in the slice (mainly four components contribution).

$$\frac{\partial n}{\partial t} A dx = \left[\frac{J_n(x) A}{-q} - \frac{J_n(x + dx) A}{-q} \right] + (G_n - R_n) A dx ,$$

Q.23 (a) Density of state of Photon : Explanation – 2 mark; Equation with explanation: 6 marks

Let us look at the integral using the number of states with a differential volume in the K-space.

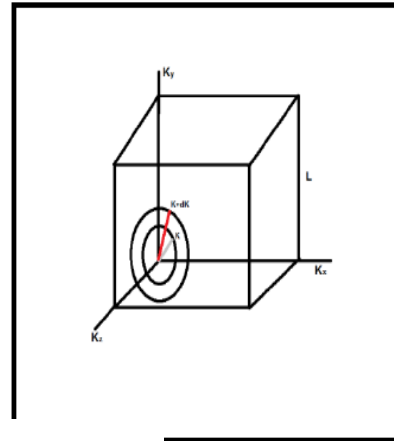
$$\frac{d^3K}{\left(\frac{2\pi}{L}\right)^3} = \frac{K^2 dk d\Omega}{\left(\frac{2\pi}{L}\right)^3}$$

Where $d\Omega$ is the differential solid angle.

Therefore

$$N(E_{21}) = \frac{2}{V} \sum_K \delta(E_2 - E_1 - E_k)$$

$$N(E_{21}) = 2 \int \frac{K^2 dk d\Omega}{(2\pi)^3} \delta(E_2 - E_1 - E_k)$$



integration over

$$K = \frac{E_k n_r}{\hbar c}$$

$$V dK = \frac{n_r 2\pi}{h c} dE_k$$

$$N(E_{21}) = 2 \int \frac{K^2}{(2\pi)^3} \frac{n_r 2\pi}{h c} dE_k (4\pi) \delta(E_{21} - E_k)$$

$$N(E_{21}) = 2 \int \frac{1}{(2\pi)^3} \left(\frac{E_k n_r 2\pi}{h c} \right)^2 \frac{n_r 2\pi}{h c} dE_k (4\pi) \delta(E_{21} - E_k)$$

$$N(E_{21}) = \frac{2 \times 4\pi \times (2\pi)^3 (n_r)^3}{(2\pi)^3 (h c)^3} \int (E_k)^3 dE_k \delta(E_{21} - E_k)$$

$$N(E_{21}) = \frac{8\pi (n_r)^3}{(h c)^3} E_{21}^2 \quad [\hbar = \frac{h}{2\pi}; h = \hbar 2\pi]$$

$$N(E_{21}) = \frac{8\pi E_{21}^2 (n_r)^3}{8\pi^3 \hbar^3 c^3} = \frac{E_{21}^2 (n_r)^3}{\pi^2 \hbar^3 c^3}$$

Which is the number of states with photon energy E_{21} per unit volume per energy interval.

(OR)

(b) (i) Concept and explanation of Fermi Golden rule :3marks

- It is also referred to as the “**Golden Rule of time-dependent perturbation theory**”.
- Fermi's golden rule is a simple expression for the transition probabilities between states of a quantum system, which are subjected to a perturbation.
- Fermi's Golden Rule provides the rate at which atomic or electronic transitions take place between two states.
- The rates are calculated from probabilities determined by transition matrix elements in quantum mechanical, first-order perturbation theory.
- It applies to a wide range of optical and electronic processes for which the initial and final states can be described by wave functions.

Equation:2 marks

- The transition probability is of the general form:

$$\lambda_{if} = \frac{2\pi}{\hbar} |M_{if}|^2 \rho_f$$

Transition probability *Matrix element for the interaction* *Density of final states*

(ii) Determine the conversion efficiency of the solar cell

if short-circuit current (I_{sc})=2.8 A, open-circuit voltage (V_{oc})=0.55V, fill factor (FF)=0.8, and input power (P_{in})=10W.

Answer: **Equation- 1mark; Calculation and Answer -2marks**

Solution:

Efficiency $\eta = \frac{I_{sc} V_{oc} FF}{P_{in}} \times 100\%$

Conversion efficiency = 12.32 %

24. (a) Hall effect : Definition of Hall effect with Diagram – 2 marks;
Relationship between Hall coefficient and Hall voltage derivation – 6 marks.

Determination of Hall coefficient

The Hall coefficient is determined by measuring the Hall voltage that generates the Hall field.

If ' w ' is the width of the sample across which the Hall voltage is measured, then

$$E_H = \frac{V_H}{w} \quad (8)$$

We know that, $R_H = \frac{E_H}{BJ} \quad (9)$

Substituting the value of E_H in the above eqn. we get,

$$R_H = \frac{V_H}{wBJ} \quad (\text{or}) \quad V_H = R_H w BJ \quad (10)$$

If the thickness of the sample is ' t ', the its cross sectional area $A = wt$, and the current density,

$$J = \frac{I}{A} = \frac{I}{wt} \quad (11)$$

Substitute the value of ' J ' in eqn., (11), we get,

$$V_H = \frac{R_H w.B.I}{wt} = \frac{R_H I B}{t} \quad (\text{or}) \quad R_H = \frac{V_H t}{IB} \quad (12)$$

V_H will be opposite in sign for P and N type semiconductors.

Experiment

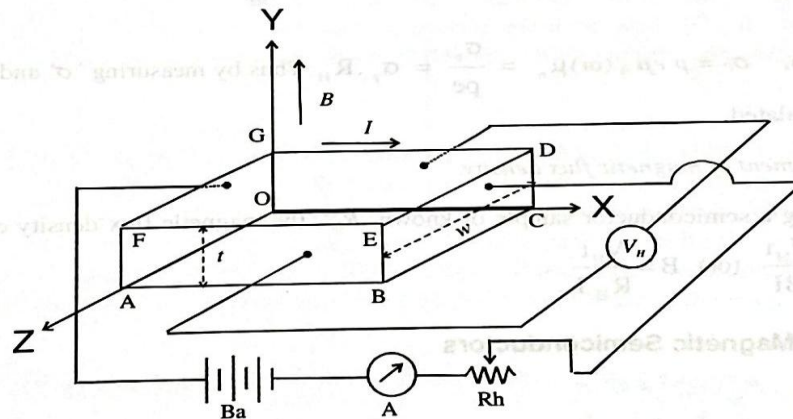


Fig.1.14. Experimental setup for the measurement of Hall voltage

A rectangular slab of the given material having thickness ' t ' and width ' w ' is taken. A current of ' I ' amperes is passed through this sample by connecting it to a battery, ' Ba '. The sample is placed between two pole pieces of an electromagnet such that the field ' B ' is perpendicular to I as shown in the experimental setup.

The hall voltage ' V_H ' is then measured by placing two probes at the two side faces of the slab. If the magnetic flux density is ' B ' and ' V_H ' is the hall voltage, then the Hall coefficient,

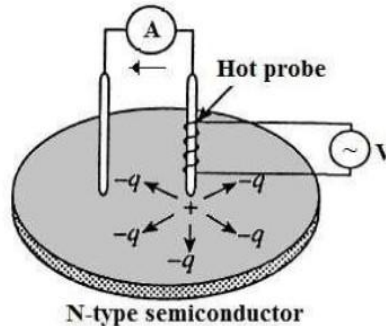
(OR)

(b) (i) **Working principle and explanation of Hot probe :3marks ; Diagram: 1 mark**

Principle:

• A conventional Hot-Probe experiment enables a simple and efficient way to distinguish between n-type and p-type semiconductors using a hot probe and a standard multi-meter.

While applying the cold and hot probes to an n-type semiconductor, positive voltage readout is obtained in the meter, whereas for a p-type semiconductor, negative voltage is obtained.



- The thermally excited majority free charged carriers are translated within the semiconductor from the hot probe to the cold probe.
- Mechanism for this motion within the semiconductor is of a diffusion type since the material is uniformly doped due to the constant heating in the hot probe contact.
- These translated majority carriers define the electrical potential sign of the measured current in the multimeter

(ii) Monte Carlo method – Definition 1` mark ;

Monte Carlo Method

Monte Carlo Simulation, also known as the Monte Carlo method or a multiple probability simulation, is a mathematical technique, which is used to estimate the possible outcome of an uncertain event. The Monte Carlo method

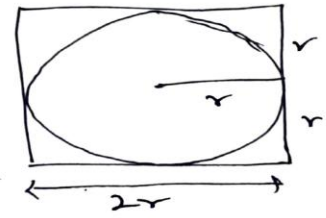
Explanation of estimation of Pi value – 3 marks

→ Consider a Square with sides of $2r$

$$S_{\text{square}} = (2r)^2 = 4r^2$$

→ Draw a circle inside the square with radius r

$$S_{\text{circle}} = \pi r^2$$



→ Thus the ratio is

$$\frac{S_{\text{circle}}}{S_{\text{square}}} = \frac{\pi r^2}{4r^2} = \frac{\pi}{4}$$

→ Multiplying 4 on the both sides of eqn

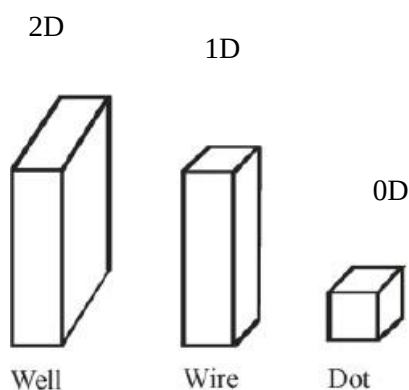
$$4 \times \frac{S_{\text{circle}}}{S_{\text{square}}} = \pi$$

$$\pi = 4 \times \frac{S_{\text{circle}}}{S_{\text{square}}}$$

→ Since

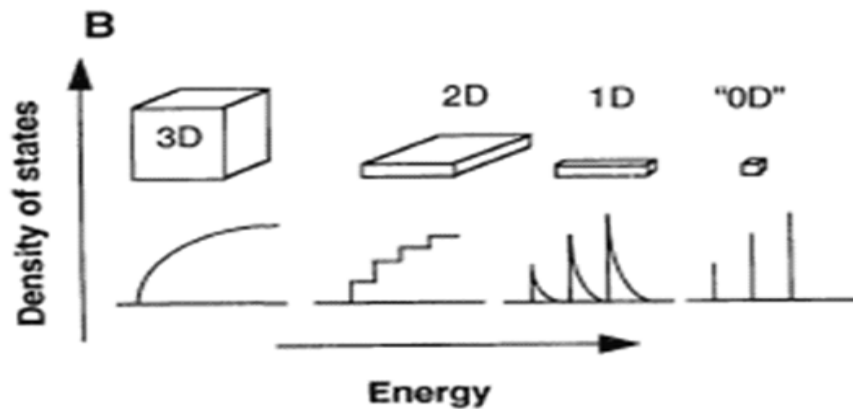
$$\pi = 4 \times \frac{S_{\text{circle}}}{S_{\text{square}}}$$

25. (a) Definition and derivation of 2D and 1D – 6 marks (Each 3 marks); 0D – explanation - 2 marks;



Quantum well, Quantum wire and Quantum dots

- When the size or dimension of a material is continuously reduced from a large or macroscopic size, such a metre or centimetre, to a very small size, the properties remain the same at first, then small changes begin to occur, until finally when the size drops below 100 nm, dramatic changes in properties can occur.
- If one dimension is reduced to the nanorange while the other dimensions remain large, then we obtain a structure known as **quantum well**.
- If two dimensions are so reduced and one remains large, the resulting structure is referred to as a **quantum wire**.
- The extreme case of this process of size reduction in which all three dimensions reach the low nanometer range is called a **quantum dot**.



2D DOS $\frac{dn}{dE} = \frac{dn}{dk} \frac{dk}{dE} = \frac{Am}{2\pi\hbar^2}$

1D DOS $\frac{dn}{dE} = \frac{Lm}{2\pi\hbar^2 k} \propto E^{-\frac{1}{2}}$

density of states of systems = $2 \times f(E - E_c)$

(OR)

25 (b) PVD – Diagram – 2 marks; Explanation – 6 Marks

- Alternative process to electroplating.
- The process is similar to chemical vapour deposition (CVD) except that the raw materials/precursors, i.e. the material that is going to be deposited starts out in solid form, whereas in CVD, the precursors are introduced to the reaction chamber in the gaseous state.

PVD processes are carried out under vacuum conditions. The process involved four steps:

1. Evaporation
2. Transportation
3. Reaction
4. Deposition

Evaporation

During this stage, a target, consisting of the material to be deposited is bombarded by a high energy source such as a beam of electrons or ions. This dislodges atoms from the surface of the target, 'vaporising' them.

Transport

This process simply consists of the movement of ‘vaporised’ atoms from the target to the substrate to be coated and will generally be a straight line affair.

Reaction

- ✚ In some cases coatings will consist of metal oxides, nitrides, carbides and other such materials.
- ✚ In these cases, the target will consist of the metal.
- ✚ The atoms of metal will then react with the appropriate gas during the transport stage.

For the above examples, the reactive gases may be oxygen, nitrogen and methane.

In instances where the coating consists of the target material alone, this step would not be part of the process

Deposition

This is the process of coating build up on the substrate surface.

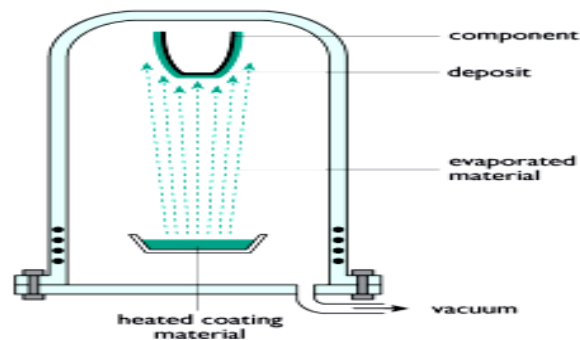
Depending on the actual process, some reactions between target materials and the reactive gases may also take place at the substrate surface simultaneously with the deposition process.

Fig. shows a schematic diagram of the principles behind one common PVD method.

The component that is to be coated is placed in a vacuum chamber. The coating material is evaporated by intense heat from, for example, **a tungsten filament.**

An alternative method is to evaporate the coating material by a complex **ion bombardment technique.**

The coating is then formed by atoms of the coating material being deposited onto the surface of the component being treated.



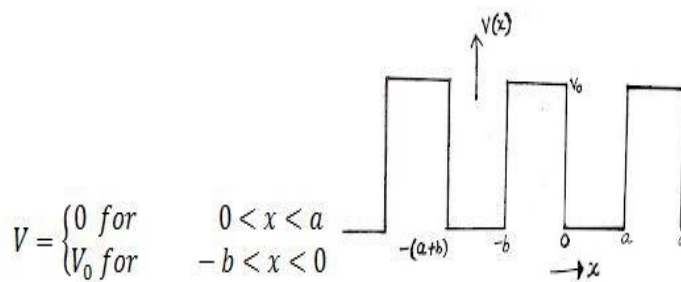
PART—C (1x15=15Marks)

Answer **ANYONE**

26 (i) Derivation for an electron in a periodic field of 1D crystal lattice- Diagram with explanation – 2 marks ; Derivation – 10 marks

According to Kronig and Penney the electrons move in a periodic square well potential.

- This potential is produced by the positive ions (ionized atoms) in the lattice.
- It is not easy to solve Schrödinger's equation with these potentials. So, Kronig and Penney approximated these potentials inside the crystal to the shape of rectangular steps as shown in Fig. (c). This model is called Kronig-Penney model of potentials.



$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

(ii) Explanation of energy of electron under p → 0 (3 marks)

Where $P = \frac{mV_0ab}{\hbar^2}$ is called the scattering power of the potential barrier.

(i) When $P = 0$:

$$\text{Eqn (13)} \Rightarrow \cos \alpha a = \cos ka$$

$$\alpha a = ka$$

$$\alpha = k$$

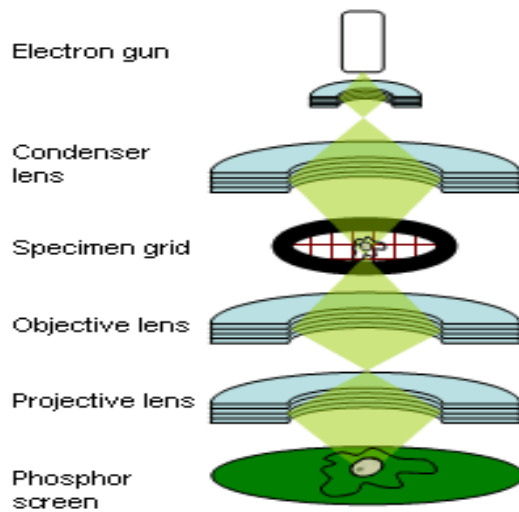
$$\alpha^2 = k^2$$

$$\frac{8\pi^2m}{h^2} E = k^2$$

$$E = \frac{\hbar^2 k^2}{2m} \quad \therefore \hbar = \frac{h}{2\pi}$$

This represents there is no band gap. so the electrons are moving freely.

27 (i) Construction working principle of TEM - 4 marks



- the [electron gun](#), produces a stream of monochromatic electrons.
- This stream is focused to a small, thin, coherent beam by the use of condenser lenses 1 and 2. The first [lens](#) (usually controlled by the "spot size knob") largely determines the "spot size"; the general size range of the final spot that strikes the sample.
- The second lens (usually controlled by the "intensity or brightness knob" actually changes the size of the spot on the sample; changing it from a wide dispersed spot to a pinpoint beam.
- The beam is restricted by the condenser [aperture](#) (usually user selectable), knocking out high angle electrons (those far from the optic axis, the dotted line down the center)
- The beam strikes the specimen and parts of it are transmitted
- This transmitted portion is focused by the objective lens into an image
- The image is passed down the column through the projector lenses, being enlarged all the way.
- The image strikes the phosphor image screen and light is generated, allowing the user to see the image

**Comment on the interaction of electron beam with sample in SEM and TEM
– 6 marks**

Specimen Interactions

TEM- Unscattered Electrons ; Elastically Scattered electrons ; Inelastically Scattered Electrons

SEM- Backscattered electrons ; Secondary electrons ; X-rays ; Auger electrons

(ii) Working concept of AFM – 4 marks ; Diagram – 1 mark

- The AFM raster scans a sharp probe over the surface of a sample and measures the changes in force between the probe tip and the sample.

Working Concept

- The physical parameter probe is a force resulting from different interactions.
- The origin of these interactions can be ionic repulsion, van der Waals, capillary, electrostatic and magnetic forces, or elastic and plastic deformations.
- Thus, **an AFM image is generated by recording the force changes as the probe (or sample) is scanned in the x , y and z directions.**
- The sample is mounted on a piezoelectric scanner, which ensures three-dimensional positioning with high resolution.
- The force is monitored by attaching the probe to a pliable cantilever, which acts as a spring, and measuring the bending or "deflection" of the cantilever.
- The larger the cantilever deflection, the higher the force that will be experienced by the probe.
- Most instruments today use an optical method to measure the cantilever deflection with high resolution; a laser beam is focused on the free end of the cantilever, and the position of the reflected beam is detected by a photodiode.
- AFM cantilevers and probes are typically made of silicon or silicon nitride(nano) by micro fabrication techniques.

